homework-5

2026-10-29

Dylan Smith Homework 5

Question 1: Simulated Data and Variable Selection

(a) Generate Predictors and Errors

```
set.seed(2024)
x <- rnorm(n = 100, mean = 0, sd = 1)
epsilon <- rnorm(n = 100, mean = 0, sd = 0.25)
```

(b) Generate the Response

```
y <- 1 + 3 * x + 2 * x^2 - x^4 + epsilon
```

(c) Exhaustive Best Subset Selection

```
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.4.3
```

```
power_data <- data.frame(
  Y = y,
  X = x,
  X2 = x^2,
  X3 = x^3,
  X4 = x^4,
  X5 = x^5,
  X6 = x^6,
  X7 = x^7,
  X8 = x^8,
  X9 = x^9,
  X10 = x^10
)

best_subsets <- regsubsets(
  Y ~ ., data = power_data, nvmax = 10,
  method = "exhaustive", really.big = TRUE
```

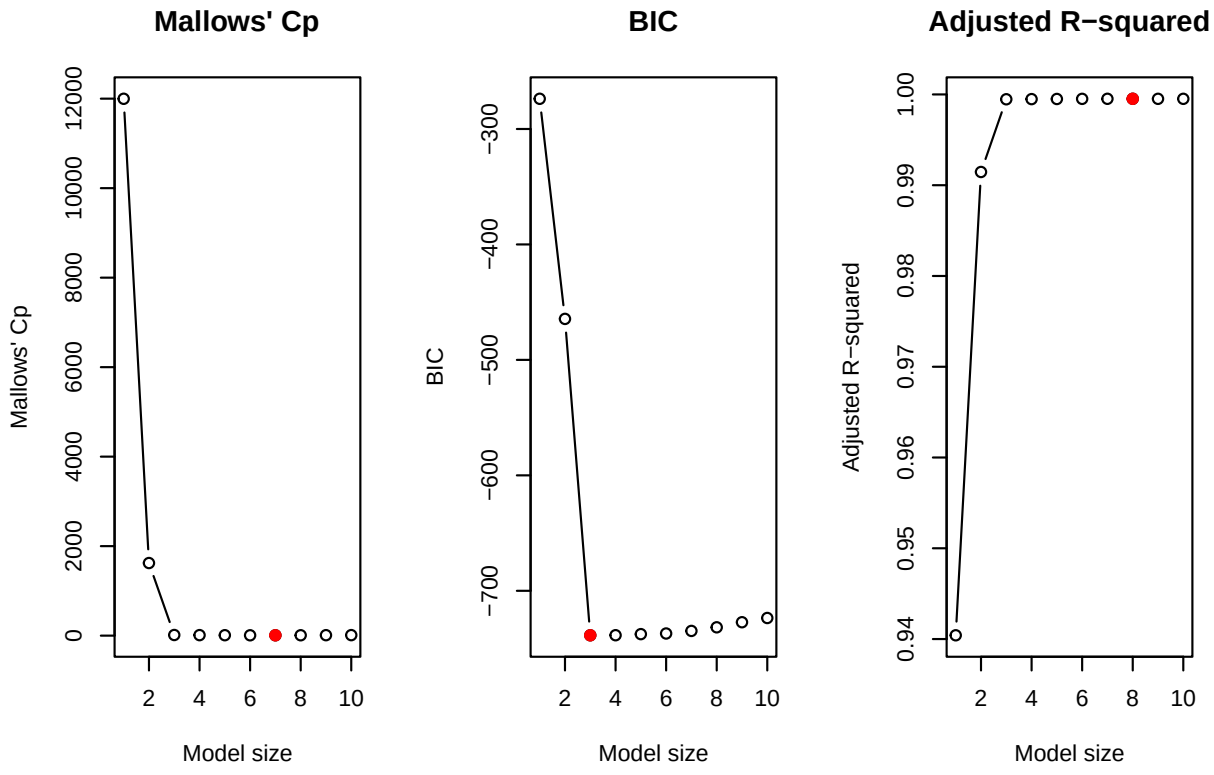
```
)
subset_summary <- summary(best_subsets)
model_sizes <- 1:10

best_cp_size <- which.min(subset_summary$cp)
best_bic_size <- which.min(subset_summary$bic)
best_adjr2_size <- which.max(subset_summary$adjr2)

par(mfrow = c(1, 3))
plot(model_sizes, subset_summary$cp, type = "b",
     xlab = "Model size", ylab = "Mallows' Cp", main = "Mallows' Cp")
points(best_cp_size, subset_summary$cp[best_cp_size], col = "red", pch = 19)

plot(model_sizes, subset_summary$bic, type = "b",
     xlab = "Model size", ylab = "BIC", main = "BIC")
points(best_bic_size, subset_summary$bic[best_bic_size], col = "red", pch = 19)

plot(model_sizes, subset_summary$adjr2, type = "b",
     xlab = "Model size", ylab = "Adjusted R-squared",
     main = "Adjusted R-squared")
points(best_adjr2_size, subset_summary$adjr2[best_adjr2_size],
       col = "red", pch = 19)
```




```
par(mfrow = c(1, 1))

report_subset <- function(criterion, model_size) {
```

```
  model_coef <- coef(best_subsets, id = model_size)
  selected_powers <- names(model_coef)[-1]
  cat("\n", criterion, " selected model (size ", model_size, ")\n", sep = "")
  cat("Selected powers: ", paste(selected_powers, collapse = ", "), "\n", sep = "")
  print(model_coef)
}

report_subset("Minimum Mallows' Cp", best_cp_size)
```

```
##
## Minimum Mallows' Cp selected model (size 7)
## Selected powers: X, X2, X4, X6, X7, X9, X10
##  (Intercept)            X           X2           X4           X6           X7
##  1.007358314  2.921733111  2.287299765 -1.336900594  0.097450463  0.028842313
##           X9          X10
## -0.006667529 -0.001813849
```

```
report_subset("Minimum BIC", best_bic_size)
```

```
##
## Minimum BIC selected model (size 3)
## Selected powers: X, X2, X4
## (Intercept)           X          X2          X4
##    1.017037    3.017862    2.024974   -1.002568
```

```
report_subset("Maximum adjusted R-squared", best_adjr2_size)
```

```
##
## Maximum adjusted R-squared selected model (size 8)
## Selected powers: X, X2, X3, X4, X5, X7, X9, X10
##   (Intercept)             X            X2            X3            X4
##  1.0275805537  3.0754550462  2.0721539703 -0.4550829045 -1.0505419857
##            X5            X7            X9           X10
##  0.4010344306 -0.1073126175  0.0089848604  0.0006758474
```

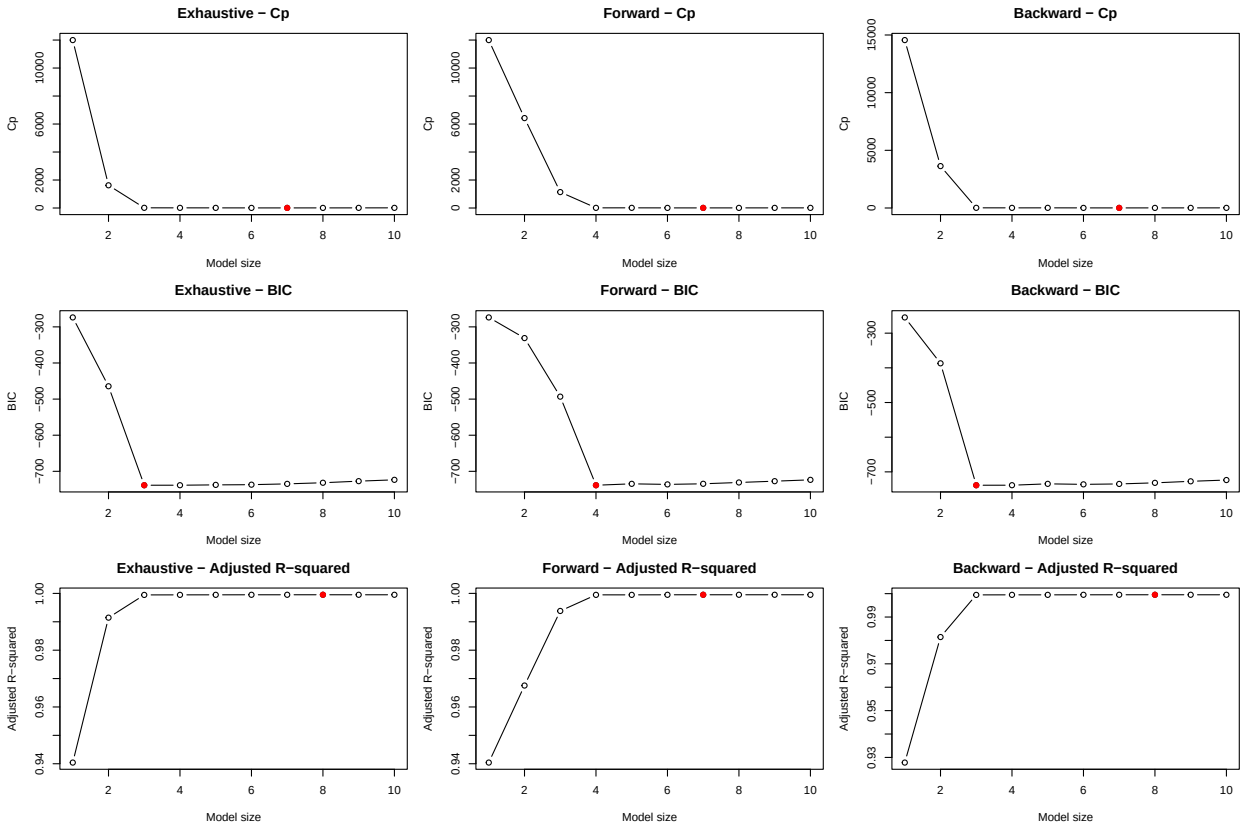
(d) Forward and Backward Selection

```
forward_subsets <- regsubsets(
  Y ~ ., data = power_data, nvmax = 10, method = "forward"
)
backward_subsets <- regsubsets(
  Y ~ ., data = power_data, nvmax = 10, method = "backward"
)

subset_fits <- list(
  Exhaustive = best_subsets,
  Forward = forward_subsets,
  Backward = backward_subsets
)
subset_summaries <- lapply(subset_fits, summary)
```

```r
criterion_names <- c("Cp", "BIC", "Adjusted R-squared")
criterion_values <- list(
  Cp = function(s) s$cp,
  BIC = function(s) s$bic,
  `Adjusted R-squared` = function(s) s$adjr2
)
criterion_optima <- lapply(subset_summaries, function(s) {
  c(Cp = which.min(s$cp), BIC = which.min(s$bic),
    `Adjusted R-squared` = which.max(s$adjr2))
})

# Plot each criterion for all three search methods, marking its optimum
par(mfrow = c(3, 3), mar = c(4, 4, 3, 1))
for (criterion in criterion_names) {
  for (method in names(subset_fits)) {
    values <- criterion_values[[criterion]](subset_summaries[[method]])
    best_size <- criterion_optima[[method]][[criterion]]
    plot(model_sizes, values, type = "b",
         xlab = "Model size", ylab = criterion,
         main = paste(method, "-", criterion))
    points(best_size, values[best_size], col = "red", pch = 19)
  }
}
```



```r
par(mfrow = c(1, 1))

# Compare selected model sizes and powers across all three search methods
selection_comparison <- do.call(rbind, lapply(names(subset_fits), function(method) {
  data.frame(
    Method = method,
    Criterion = criterion_names,
    Model_Size = as.integer(criterion_optima[[method]][criterion_names]),
    Selected_Powers = vapply(criterion_names, function(criterion) {
      model_coef <- coef(
        subset_fits[[method]],
        id = criterion_optima[[method]][[criterion]]
      )
      paste(names(model_coef)[-1], collapse = ", ")
    }, character(1)),
    row.names = NULL
  )
}))
print(selection_comparison, row.names = FALSE)
```

```
##      Method          Criterion Model_Size                Selected_Powers
##  Exhaustive                 Cp          7     X, X2, X4, X6, X7, X9, X10
##  Exhaustive                BIC          3                       X, X2, X4
##  Exhaustive Adjusted R-squared          8 X, X2, X3, X4, X5, X7, X9, X10
##     Forward                 Cp          7     X, X2, X4, X5, X8, X9, X10
##     Forward                BIC          4                  X, X2, X4, X5
##     Forward Adjusted R-squared          7     X, X2, X4, X5, X8, X9, X10
##    Backward                 Cp          7      X, X2, X3, X4, X5, X7, X9
##    Backward                BIC          3                       X, X2, X4
##    Backward Adjusted R-squared          8 X, X2, X3, X4, X5, X7, X9, X10
```

```r
# Print the coefficient estimates for every selected model
for (method in names(subset_fits)) {
  for (criterion in criterion_names) {
    model_size <- criterion_optima[[method]][[criterion]]
    cat("\n", method, " selection by ", criterion,
        " (model size ", model_size, ")\n", sep = "")
    print(coef(subset_fits[[method]], id = model_size))
  }
}
```

```
##
## Exhaustive selection by Cp (model size 7)
##  (Intercept)            X           X2           X4           X6           X7
##  1.007358314  2.921733111  2.287299765 -1.336900594  0.097450463  0.028842313
##           X9          X10
## -0.006667529 -0.001813849
##
## Exhaustive selection by BIC (model size 3)
## (Intercept)           X          X2          X4
##    1.017037    3.017862    2.024974   -1.002568
##
## Exhaustive selection by Adjusted R-squared (model size 8)
```

```
##   (Intercept)             X            X2            X3            X4
##  1.0275805537  3.0754550462  2.0721539703 -0.4550829045 -1.0505419857
##            X5            X7            X9           X10
##  0.4010344306 -0.1073126175  0.0089848604  0.0006758474
##
## Forward selection by Cp (model size 7)
##  (Intercept)            X           X2           X4           X5           X8
##  1.021903248  2.907599652  2.138554481 -1.117288665  0.044485684  0.012038450
##           X9          X10
## -0.002079225 -0.001558401
##
## Forward selection by BIC (model size 4)
##  (Intercept)            X           X2           X4           X5
##  1.049596469  2.974968105  1.924519402 -0.969179053  0.007936094
##
## Forward selection by Adjusted R-squared (model size 7)
##  (Intercept)            X           X2           X4           X5           X8
##  1.021903248  2.907599652  2.138554481 -1.117288665  0.044485684  0.012038450
##           X9          X10
## -0.002079225 -0.001558401
##
## Backward selection by Cp (model size 7)
##  (Intercept)            X           X2           X3           X4           X5
##  1.046986401  3.093068231  1.960607266 -0.473889766 -0.985931089  0.386259362
##           X7           X9
## -0.088836582  0.005335541
##
## Backward selection by BIC (model size 3)
## (Intercept)           X          X2          X4
##    1.017037    3.017862    2.024974   -1.002568
##
## Backward selection by Adjusted R-squared (model size 8)
##   (Intercept)             X            X2            X3            X4
##  1.0275805537  3.0754550462  2.0721539703 -0.4550829045 -1.0505419857
##            X5            X7            X9           X10
##  0.4010344306 -0.1073126175  0.0089848604  0.0006758474
```

(e) Lasso with Cross-Validation

(f) Repeat the Analysis for the Fifth-Power Model

Question 2: Predicting College Applications

(a) Training/Test Split and Cross-Validation Folds

(b) Ordinary Least Squares

(c) Ridge Regression

(d) Lasso Regression

(e) Principal Components Regression

(f) Partial Least Squares

(g) Compare Test Errors with the Intercept-Only Benchmark

Question 3: Predicting Boston Crime Rates

(a) Compare Prediction Methods Using Cross-Validation

(b) Recommend a Method and Assess Test Performance

(c) Predictors in the Recommended Model